

Extending Random Survival Forests to Left-Truncated Right-Censored Data

Project outline

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1 Motivation

Random survival forests (RSF) extend Breiman's random forests to right-censored survival data by growing survival trees, using survival-based splitting rules, and estimating a terminal-node cumulative hazard function. The standard RSF framework assumes that each individual is at risk from time zero until failure or censoring. In many biomedical cohort studies this assumption is inappropriate because individuals enter the risk set after time zero. This is the setting of *left truncation*, also called *delayed entry*.

The goal of this project is to extend the RSF algorithm to data with both left truncation and right censoring. The main idea is simple: wherever the standard RSF uses the right-censored risk set

$$Y_i(t) = \mathbb{1}(\tilde{T}_i \geq t),$$

we replace it by the delayed-entry risk set

$$Y_i^{LT}(t) = \mathbb{1}(L_i \leq t \leq \tilde{T}_i),$$

where L_i is the entry time. This correction must be applied consistently in the splitting rule, the terminal-node cumulative hazard estimator, and the out-of-bag prediction-error calculations.

2 Observed data and assumptions

For subject $i = 1, \dots, n$, the observed data are

$$O_i = (L_i, \tilde{T}_i, \delta_i, X_i),$$

where

- L_i is the left-truncation or delayed-entry time;
- T_i is the event time;
- C_i is the right-censoring time;
- $\tilde{T}_i = \min(T_i, C_i)$ is the observed follow-up time;
- $\delta_i = \mathbb{1}(T_i \leq C_i)$ is the event indicator;
- $X_i \in \mathbb{R}^p$ is the covariate vector.

Only subjects satisfying

$$L_i \leq \tilde{T}_i$$

are observed. The basic independent truncation and censoring assumptions are

$$T_i \perp L_i \mid X_i, \quad T_i \perp C_i \mid X_i, L_i,$$

with suitable positivity conditions, namely

$$P(L_i \leq t \leq C_i \mid X_i) > 0$$

over the time region in which survival is to be estimated.

3 Background and relation to existing work

Random survival forests (RSF), as introduced by Ishwaran et al., extend Breiman’s random forests to right-censored survival data. The key components of the RSF construction are: bootstrap sampling of subjects, random selection of candidate variables at each node, survival-based splitting rules, terminal-node cumulative hazard estimation, and ensemble averaging of cumulative hazard functions. In the classical right-censored RSF, the splitting rule is typically based on a log-rank-type statistic, and terminal-node prediction is based on the Nelson–Aalen estimator computed within each terminal node.

The standard RSF formulation is designed for right-censored data, where the at-risk process is

$$Y_i(t) = \mathbf{1}(\tilde{T}_i \geq t),$$

with $\tilde{T}_i = \min(T_i, C_i)$. This definition is no longer valid in the presence of left truncation, also known as delayed entry. If subject i enters the risk set only at time L_i , then the correct at-risk process is

$$Y_i^{LT}(t) = \mathbf{1}(L_i \leq t \leq \tilde{T}_i).$$

Thus, a direct application of ordinary RSF to left-truncated data uses incorrect risk sets, and may lead to biased splitting, biased terminal-node survival estimates, and biased prediction-error evaluation.

Several tree and forest methods for left-truncated/right-censored (LTRC) data already exist . . .

4 Proposed contribution

The goal of this project is to develop a transparent Ishwaran-style random survival forest for left-truncated/right-censored data. The central idea is simple but fundamental: every place in the RSF algorithm where a risk set is used must be modified to respect delayed entry. Specifically, we replace

$$Y_i(t) = \mathbf{1}(\tilde{T}_i \geq t)$$

by

$$Y_i^{LT}(t) = \mathbf{1}(L_i \leq t \leq \tilde{T}_i).$$

This replacement affects four parts of the algorithm:

splitting, terminal-node estimation, ensemble prediction, out-of-bag validation.

First, at each candidate split, we propose to use a delayed-entry log-rank statistic. For a candidate split into left and right daughter nodes, define

$$Y_L^{LT}(t) = \sum_{i \in L} \mathbf{1}(L_i \leq t \leq \tilde{T}_i), \quad Y_R^{LT}(t) = \sum_{i \in R} \mathbf{1}(L_i \leq t \leq \tilde{T}_i),$$

and

$$Y^{LT}(t) = Y_L^{LT}(t) + Y_R^{LT}(t).$$

At an event time t_j , let

$$d_L(t_j) = \sum_{i \in L} \mathbf{1}(\tilde{T}_i = t_j, \delta_i = 1), \quad d_R(t_j) = \sum_{i \in R} \mathbf{1}(\tilde{T}_i = t_j, \delta_i = 1),$$

and $d(t_j) = d_L(t_j) + d_R(t_j)$. The delayed-entry log-rank numerator is

$$U_{LT} = \sum_{t_j} \left[d_L(t_j) - d(t_j) \frac{Y_L^{LT}(t_j)}{Y^{LT}(t_j)} \right],$$

with variance

$$V_{LT} = \sum_{t_j} \frac{Y_L^{LT}(t_j) Y_R^{LT}(t_j) d(t_j) \{Y^{LT}(t_j) - d(t_j)\}}{\{Y^{LT}(t_j)\}^2 \{Y^{LT}(t_j) - 1\}}.$$

The split statistic is

$$Z_{LT} = \frac{U_{LT}}{\sqrt{V_{LT}}},$$

and the selected split is the one maximizing $|Z_{LT}|$ over the candidate variables and split points.

Second, after a tree is grown, terminal-node prediction will use the delayed-entry Nelson–Aalen estimator. For a terminal node h , define

$$Y_h^{LT}(t) = \sum_{i \in h} \mathbf{1}(L_i \leq t \leq \tilde{T}_i),$$

and

$$d_h(t_j) = \sum_{i \in h} \mathbf{1}(\tilde{T}_i = t_j, \delta_i = 1).$$

The terminal-node cumulative hazard estimator is

$$\hat{H}_h^{LT}(t) = \sum_{t_j \leq t} \frac{d_h(t_j)}{Y_h^{LT}(t_j)}.$$

The corresponding survival curve may be estimated either by

$$\hat{S}_h^{LT}(t) = \exp\{-\hat{H}_h^{LT}(t)\},$$

or by the delayed-entry Kaplan–Meier estimator

$$\hat{S}_h^{LT}(t) = \prod_{t_j \leq t} \left\{ 1 - \frac{d_h(t_j)}{Y_h^{LT}(t_j)} \right\}.$$

Third, ensemble prediction will follow the original RSF logic. A new subject with covariate vector x is dropped down each tree. The tree-specific prediction is the delayed-entry Nelson–Aalen

cumulative hazard of the terminal node into which x falls. Averaging over trees gives the ensemble cumulative hazard function

$$\hat{H}_e^{LT}(t | x) = \frac{1}{B} \sum_{b=1}^B \hat{H}_b^{LT}(t | x).$$

Similarly, an ensemble survival curve can be obtained from the corresponding tree-specific or ensemble cumulative hazard estimates.

Fourth, out-of-bag prediction error must also respect left truncation. The ordinary right-censored C-index and Brier score are not automatically valid when subjects enter the risk set at different delayed-entry times. Therefore, the project will implement and compare prediction-error estimators that use the correct left-truncated risk sets or appropriate inverse-probability adjustments for delayed-entry sampling.

5 Standard RSF components that require modification

The original RSF algorithm for right-censored data includes the following core components:

1. draw bootstrap samples of subjects;
2. grow survival trees using a survival splitting rule, commonly a log-rank splitting rule;
3. stop splitting when terminal nodes are sufficiently small or contain too few events;
4. estimate a cumulative hazard function within each terminal node using a Nelson–Aalen estimator;
5. average tree-specific cumulative hazard functions to obtain the ensemble cumulative hazard;
6. estimate prediction error using out-of-bag observations.

For left-truncated data, steps 2, 4, and 6 must be modified. Step 1 can still resample subjects, but all risk sets inside each bootstrap tree must be recomputed using the in-bag subjects and their entry times L_i .

6 Left-truncation-adjusted log-rank splitting rule

Consider a node $\mathcal{N}$ containing a set of subjects. For a candidate split based on variable $X^{(m)}$ and cutpoint c , define the left and right daughter nodes

$$\mathcal{N}_L = \{i \in \mathcal{N} : X_i^{(m)} \leq c\}, \quad \mathcal{N}_R = \{i \in \mathcal{N} : X_i^{(m)} > c\}.$$

Let $t_1 < t_2 < \dots < t_K$ denote the distinct observed event times in the parent node. For each event time t_k , define the delayed-entry risk sets

$$Y_L(t_k) = \sum_{i \in \mathcal{N}_L} \mathbb{1}(L_i \leq t_k \leq \tilde{T}_i),$$

$$Y_R(t_k) = \sum_{i \in \mathcal{N}_R} \mathbb{1}(L_i \leq t_k \leq \tilde{T}_i),$$

with

$$Y(t_k) = Y_L(t_k) + Y_R(t_k).$$

Similarly, define the number of events at t_k in the two daughter nodes,

$$d_L(t_k) = \sum_{i \in \mathcal{N}_L} \mathbb{1}(\tilde{T}_i = t_k, \delta_i = 1),$$

$$d_R(t_k) = \sum_{i \in \mathcal{N}_R} \mathbb{1}(\tilde{T}_i = t_k, \delta_i = 1),$$

and

$$d(t_k) = d_L(t_k) + d_R(t_k).$$

The left-truncation-adjusted log-rank numerator is

$$U_{LT}(X^{(m)}, c) = \sum_{k=1}^K \left\{ d_L(t_k) - d(t_k) \frac{Y_L(t_k)}{Y(t_k)} \right\}.$$

The corresponding variance estimator is

$$V_{LT}(X^{(m)}, c) = \sum_{k=1}^K \frac{Y_L(t_k) Y_R(t_k) d(t_k) \{Y(t_k) - d(t_k)\}}{Y(t_k)^2 \{Y(t_k) - 1\}},$$

where terms with $Y(t_k) \leq 1$ are omitted. The splitting statistic is

$$Z_{LT}(X^{(m)}, c) = \frac{U_{LT}(X^{(m)}, c)}{\sqrt{V_{LT}(X^{(m)}, c)}}.$$

The best split is selected by maximizing

$$|Z_{LT}(X^{(m)}, c)|$$

over candidate variables and candidate cutpoints.

6.1 Weighted log-rank variant

A weighted version can also be used:

$$U_{LT,w}(X^{(m)}, c) = \sum_{k=1}^K w(t_k) \left\{ d_L(t_k) - d(t_k) \frac{Y_L(t_k)}{Y(t_k)} \right\},$$

with variance

$$V_{LT,w}(X^{(m)}, c) = \sum_{k=1}^K w(t_k)^2 \frac{Y_L(t_k) Y_R(t_k) d(t_k) \{Y(t_k) - d(t_k)\}}{Y(t_k)^2 \{Y(t_k) - 1\}}.$$

7 Terminal-node cumulative hazard and survival estimation

After a tree is grown, each terminal node h contains a subset of subjects. Let $t_{1,h} < \dots < t_{K_h,h}$ denote the distinct event times in terminal node h . Define

$$d_h(t_k) = \sum_{i \in h} \mathbb{1}(\tilde{T}_i = t_k, \delta_i = 1),$$

and the left-truncation-adjusted number at risk

$$Y_h^{LT}(t_k) = \sum_{i \in h} \mathbb{1}(L_i \leq t_k \leq \tilde{T}_i).$$

The terminal-node Nelson–Aalen estimator becomes

$$\hat{H}_h^{LT}(t) = \sum_{t_k, h \leq t} \frac{d_h(t_k, h)}{Y_h^{LT}(t_k, h)}.$$

The corresponding Kaplan–Meier product-limit estimator is

$$\hat{S}_h^{LT}(t) = \prod_{t_k, h \leq t} \left\{ 1 - \frac{d_h(t_k, h)}{Y_h^{LT}(t_k, h)} \right\}.$$

For a new subject with covariates x , drop x down the tree to a terminal node $h_b(x)$ in tree b . The tree-specific cumulative hazard prediction is

$$\hat{H}_b^{LT}(t \mid x) = \hat{H}_{h_b(x)}^{LT}(t).$$

The ensemble cumulative hazard function is

$$\hat{H}_e^{LT}(t \mid x) = \frac{1}{B} \sum_{b=1}^B \hat{H}_b^{LT}(t \mid x),$$

and the corresponding survival prediction can be defined as

$$\hat{S}_e^{LT}(t \mid x) = \exp\{-\hat{H}_e^{LT}(t \mid x)\}.$$

8 Out-of-bag prediction

Let $I_{ib} = 1$ if subject i is out-of-bag for tree b , and $I_{ib} = 0$ otherwise. The OOB ensemble cumulative hazard for subject i is

$$\hat{H}_{e,OOB}^{LT}(t \mid X_i) = \frac{\sum_{b=1}^B I_{ib} \hat{H}_b^{LT}(t \mid X_i)}{\sum_{b=1}^B I_{ib}}.$$

The OOB survival prediction is

$$\hat{S}_{e,OOB}^{LT}(t \mid X_i) = \exp\{-\hat{H}_{e,OOB}^{LT}(t \mid X_i)\}.$$

For risk ranking, one may use an ensemble mortality score such as

$$\hat{M}_i^{LT}(\tau) = \sum_{t_j \leq \tau} \hat{H}_{e,OOB}^{LT}(t_j \mid X_i),$$

or the predicted restricted mean survival time

$$\widehat{RMST}_i(\tau) = \int_0^\tau \hat{S}_{e,OOB}^{LT}(t \mid X_i) dt.$$

9 Prediction-error evaluation under left truncation

Prediction-error evaluation must also respect delayed entry. Naive survival C-index or Brier score calculations may be biased because the validation data are observed conditional on surviving to entry.

9.1 Left-truncation-aware concordance

For two subjects i and j , a comparable pair should satisfy that the earlier event time is observable for the other subject. A simple delayed-entry restriction is:

$$\max(L_i, L_j) \leq \min(\tilde{T}_i, \tilde{T}_j).$$

For event-time ordering, if $\tilde{T}_i < \tilde{T}_j$ and $\delta_i = 1$, then subject j must have entered before $\tilde{T}_i$:

$$L_j \leq \tilde{T}_i.$$

Compare several definitions and justify the selected concordance estimator.

9.2 Left-truncation-aware Brier score

At a fixed time t , a Brier score can be based on

$$\left\{ \mathbb{1}(T_i > t) - \hat{S}(t \mid X_i) \right\}^2,$$

with inverse probability of censoring weights and conditioning on being under observation at the relevant time. A practical version should restrict to subjects with

$$L_i \leq t$$

and apply censoring weights estimated with delayed-entry risk sets.

10 Algorithm

Algorithm 1 Left-truncated random survival forest

Require: Data $(L_i, \tilde{T}_i, \delta_i, X_i)$, $i = 1, \dots, n$; number of trees B ; number of candidate variables $mtry$; terminal-node event threshold d_0 .

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1: for  $b = 1, \dots, B$  do
2:   Draw a bootstrap sample of subjects.
3:   Start with the root node containing the in-bag subjects.
4:   while a node can be split do
5:     Randomly select  $mtry$  candidate variables.
6:     for each candidate variable and cutpoint do
7:       Compute  $Y_L(t), Y_R(t), d_L(t), d_R(t)$  using delayed-entry risk sets.
8:       Compute  $Z_{LT}$ .
9:     end for
10:    Split the node using the variable and cutpoint maximizing  $|Z_{LT}|$ .
11:  end while
12:  In every terminal node, compute  $\hat{H}_h^{LT}(t)$  using the delayed-entry Nelson–Aalen estimator.
13: end for
14: Average tree-specific cumulative hazard functions to obtain  $\hat{H}_e^{LT}(t | x)$ .
15: Use OOB predictions and left-truncation-aware performance metrics for validation.
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11 Simulation study

Compare:

1. standard RSF that ignores left truncation;
2. the proposed left-truncated RSF;
3. Cox proportional hazards with left truncation;

11.1 Performance measures

Report:

- integrated Brier score with delayed-entry correction;
- left-truncation-aware C-index;
- calibration of predicted survival at fixed times;
- bias and variance of predicted survival curves in selected covariate strata;
- variable-selection performance if variable importance is implemented.

12 Expected deliverables

The final project should include:

1. a clear mathematical description of left-truncated RSF;

2. implementation of the proposed splitting rule and terminal-node estimator;
3. simulation comparing naive RSF and left-truncated RSF;
4. at least one real-data illustration;
5. a manuscript-style report with algorithm, results, figures, and discussion;
6. reproducible code.

13 Possible extensions

If time permits, we can consider:

- Time-sensitive splitting rule for non-proportional hazards - described below;
- competing risks with delayed entry, using Aalen–Johansen terminal-node estimators;
- time-dependent covariates ???;
- variable importance and minimal depth under delayed entry;

14 Time-sensitive splitting for non-proportional hazards

The original random survival forest splitting rule is typically based on a log-rank statistic. For a candidate split of a node into left and right daughter nodes, let

$$Y_L(t), \quad Y_R(t)$$

denote the numbers at risk in the two daughter nodes at time t , and let

$$Y(t) = Y_L(t) + Y_R(t).$$

At an observed event time t_j , let

$$d_L(t_j), \quad d_R(t_j)$$

denote the numbers of events in the two daughter nodes, and let

$$d(t_j) = d_L(t_j) + d_R(t_j).$$

The usual log-rank numerator accumulated up to time t is

$$U(t) = \sum_{t_j \leq t} \left[d_L(t_j) - d(t_j) \frac{Y_L(t_j)}{Y(t_j)} \right].$$

Its variance estimate is

$$\hat{V}(t) = \sum_{t_j \leq t} \frac{Y_L(t_j)Y_R(t_j)d(t_j)\{Y(t_j) - d(t_j)\}}{Y(t_j)^2\{Y(t_j) - 1\}}.$$

The standardized cumulative log-rank process is therefore

$$Z(t) = \frac{U(t)}{\sqrt{\hat{V}(t)}}.$$

The standard log-rank splitting rule uses only the final value of this process,

$$Q_{\text{LR}} = |Z(\tau)|,$$

where τ is the largest event time considered in the node. This rule is well suited to proportional-hazards-type alternatives, where the survival difference between daughter nodes accumulates in a stable direction over follow-up.

However, when covariate effects vary with time, the cumulative log-rank process may show a large deviation during part of follow-up but then return toward zero. For example, an early harmful effect followed by no later effect, or crossing hazards, can lead to cancellation in the final statistic $Z(\tau)$. In such cases, the ordinary log-rank split may fail to select an important covariate.

We therefore propose a supremum log-rank splitting rule. For each candidate split, define

$$Q_{\text{supLR}} = \sup_{0 < t \leq \tau} |Z(t)|.$$

In practice, since $Z(t)$ changes only at observed event times, this is computed as

$$Q_{\text{supLR}} = \max_{t_j \leq \tau} \left| \frac{\sum_{t_\ell \leq t_j} \left[d_L(t_\ell) - d(t_\ell) \frac{Y_L(t_\ell)}{Y(t_\ell)} \right]}{\left[\sum_{t_\ell \leq t_j} \frac{Y_L(t_\ell) Y_R(t_\ell) d(t_\ell) \{Y(t_\ell) - d(t_\ell)\}}{Y(t_\ell)^2 \{Y(t_\ell) - 1\}} \right]^{1/2}} \right|.$$

The selected split is the candidate split that maximizes

$$Q_{\text{supLR}}.$$

This modification preserves the original random survival forest structure: trees are still grown recursively, terminal-node prediction is still based on the Nelson–Aalen estimator, and ensemble prediction is obtained by averaging tree-specific cumulative hazard functions. The only change is the splitting criterion. Instead of selecting splits that maximize the final log-rank separation, the proposed rule selects splits that maximize the largest standardized survival separation over follow-up.

The advantage is that no user-specified time intervals are required. The method automatically detects whether the separation occurs early, late, or in the middle of follow-up. Thus, it is designed to improve split selection under non-proportional hazards and time-varying covariate effects while remaining close to the original Ishwaran-style random survival forest algorithm.

What exists is related, but different:

1. Standard RSF implementations, such as randomForestSRC and scikit-survival, use the ordinary log-rank splitting rule, essentially based on the final accumulated statistic. The randomForestSRC documentation describes `splitrule="logrank"` as the default survival splitting rule, and scikit-survival also states that split quality is measured by the log-rank splitting rule.
2. Weighted log-rank survival trees exist. These allow the split criterion to emphasize early or late events by using time weights. This is related to our motivation, but it requires choosing or tuning the weights, whereas the supremum process avoids prespecified time windows or weights. (IuliiVasilev, MikhailPetrovskiy, IgorMashechkin)
3. There is work showing that log-rank-based survival forests can have low power under non-proportional hazards, and that alternative splitting procedures may help. This supports our motivation, but it is not the same as the specific supremum log-rank process split. (Natalia Korepanova, Heidi Seibold, Verena Steffen, Torsten Hothorn)

4. There are maximally selected rank statistic forests, but their goal is mainly unbiased split-variable selection and split-point selection, not specifically detecting time-local or crossing survival differences through the cumulative log-rank process. (Marvin N. Wright, Theresa Dankowski, Andreas Ziegler)

To our knowledge, standard random survival forest implementations do not use a supremum-type cumulative log-rank-process splitting rule. Existing methods mainly use ordinary log-rank splitting, log-rank-score variants, weighted log-rank splitting, conditional-inference splitting, or maximally selected rank statistics. The proposed split criterion differs by using the entire cumulative log-rank process and selecting the split with the largest standardized deviation over follow-up, thereby targeting non-proportional and time-varying effects without requiring user-specified time intervals or prespecified weights.

References

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